

Reason about gradients

AICE3002 / AICE6001

Differentiation and Optimisation Problem Class

Shapes, computation graphs, mode choice, and diagnosis

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How the problem class will work

Reading shapes from expressions

Constructing Jacobians

Computing Jacobian–vector products

Computing vector–Jacobian products

Tracing a computational graph

Choosing forward or reverse mode

Checking a gradient numerically

Diagnosing an optimisation trace

Bringing the gradient story together